

# BCSE204L Design and Analysis of Algorithms

Module-4: Topic-3: Flow Networks-Ford  
Fulkerson Algorithm

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# Flow network basics

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# Flow network definition

- A flow network  $G(V, E)$  is a directed graph where each edge  $(u, v) \in E$  has a capacity  $C(u, v) \geq 0$ , a source vertex  $s$ , and a sink vertex  $t$
- A flow  $f$  in  $G$  is a real valued function  $f : V \times V \rightarrow \mathbb{R}$  such that:
  1. Capacity constraint:-
$$f(u, v) \leq C(u, v)$$
  2. Skew symmetry:-
$$f(u, v) = -f(v, u)$$
  3. Flow conservation:-
$$\forall v \notin \{s, t\}, \sum_{u \in V} f(u, v) = \sum_{w \in V} f(v, w)$$

# A Flow network and Flow assignment

Draw a sample flow network and checking all rules of a valid flow are satisfied (on board)

# Optimization problem

- Given a network, what is the maximum flow that can be sent from source to sink
- Maximizing the amount of flow
- Observe that the total outgoing flow from  $s$  equals total incoming flow to  $t$ , due to flow conservation rule
- Formally, the value of a flow  $f$  is
$$|f| = \sum_{u \in V} f(s, u) = \sum_{u \in V} f(u, t)$$
- To solve this problem, we need to define
  1. Residual network
  2. Augmenting path

# Ford Fulkerson algorithm

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# Residual network

Consider an arbitrary flow  $f$  in a network  $G$ . The residual network  $G_f$  has the same vertices as the original network, and exactly one or two edges for each edge  $(u, v)$  in the original network based on

1. if  $f(u, v) < C(u, v)$  then there is a forward edge  $(u, v)$  with capacity  $C_f(u, v) = C(u, v) - f(u, v)$
2. if  $f(u, v) > 0$  then there is a backward edge  $(v, u)$  with capacity  $C_f(v, u) = f(u, v)$

# Residual network example

On board

# Augmenting path

- An augmenting path in a residual network is a path from source  $s$  to sink  $t$ , whose purpose is to increase (augment) the flow
- The path capacity  $\delta(p)$  of a path  $p$  is the minimum capacity of the edge along the path

# Ford Fulkerson algorithm

**procedure** FORD-FULKERSON( $G, s$ )

**Input:** flow network  $G$

**Output:** maximum flow  $f$

**while** there exist an augmenting path in  $G_f$  **do**

    identify an augmenting  $p$  in  $G_f$

$\delta(p) = \min\{C_f(u, v) \mid (u, v) \in p\}$

    augment  $\delta(p)$  units of flow along  $p$

    update  $G_f$

**end while**

    recover maximum flow from the final  $G_f$

**end procedure**

# Ford Fulkerson Analysis

- Time complexity of Ford Fulkerson algorithm is  $O(|f| * E)$ ,  $|f|$  is the max flow.
- We run a loop while there is an augmenting path
- In worst case, we may add 1 unit flow in every iteration
- The maximum number of edges between  $s$  and  $t$  can be  $|E|$
- Therefore the time complexity becomes  $O(|f| * E)$

# Max-flow and min-cut

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# Cut

- A cut  $(S, T)$  in a flow network  $G(V, E)$  partitions the vertex set  $V$  into two subsets  $S \subset V$  and  $T = V - S$ , and it consists of edges with one endpoint in  $S$  and the other in  $T$
- The cut  $(S, T)$  is an  $s - t$  cut if  $s \in S$  and  $t \in T$
- The capacity of an  $s - t$  cut is
$$C(S, T) = \sum_{u \in S, v \in T} C(u, v)$$
- A min-cut is  $s - t$  cut having the minimum capacity

# Max-flow min-cut theorem

- If  $f$  is a flow in a flow network  $G$  with source  $s$  and sink  $t$  then the following are equivalent.
  1.  $f$  is a max-flow in  $G$
  2. The residual network  $G_f$  contains no augmenting path
  3.  $|f| = C(S, T)$  for some cut  $(S, T)$
- This acts as proof for Ford Fulkerson algorithm
- To get the cut specified in (3) above (the min-cut) construct  $S$  with all vertices that are reachable from the source vertex  $s$  in the residual network  $G_f$  and  $T$  with the remaining vertices.
- Alternatively
$$S = \{v \in V \mid \text{there exist a path from } s \text{ to } v \text{ in } G_f\}$$
- and  $T = V - S$
- Thus,  $(S, T)$  is as  $S - T$  cut since  $s \in S$  and  $t \in T$

# Important observations

- For each edge  $(u, v) \in (S, T)$ , we have  $f(u, v) = C(u, v)$
- For each edge  $(v, u) \in (T, S)$ , we have  $f(u, v) = 0$

# Illustrative proof

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On board

Thank You